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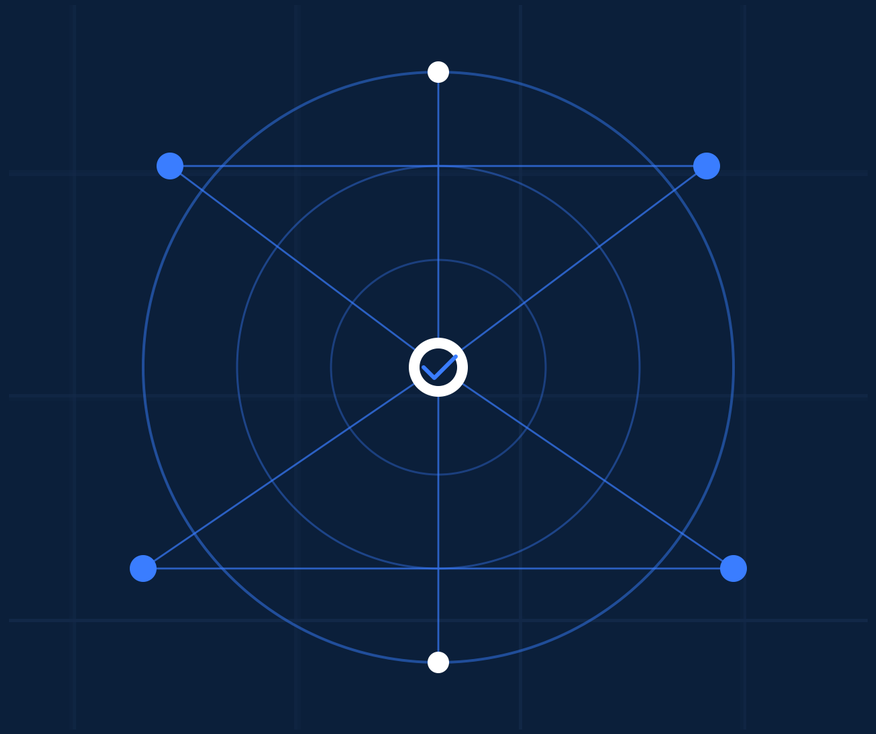

A LAYERED PROOF OF RESERVES FRAMEWORK

LPOR

Usable and Publicly Auditable Solvency Verification

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Public Auditability · Layered Verification · Solvency

When trust fails at global scale

Most everyday users still access crypto through custodians — and custodial failures have occurred throughout the industry's growth.

GROWTH OF THE CUSTODIAL ERA

CUSTODIAL FAILURES AT EACH STAGE

— pre-custodial era —

2014

Mt. Gox

~850,000 BTC lost

Largest early-era exchange failure. No verifiable reserve disclosure.

2022

FTX

~$8B customer shortfall

Liabilities misrepresented to users despite a regulated appearance.

2026

Bithumb

~620,000 BTC mis-credited

User balances exceeded actual reserves — a real-time liability mismatch.

THE MOTIVATING QUESTION

How can users actually know that an exchange holds their funds?

The evolution of Proof of Reserves

Three generations, each cryptographically more sophisticated than the last.

GENERATION 1

Wallet Disclosure

Assets only.
No liability accounting.

Assets visible on-chain

GENERATION 2

Merkle PoR

Liability inclusion.
Per-user Merkle proofs.

Liabilities + assets compared

GENERATION 3

zk-PoR

Cryptographic solvency.
Zero-knowledge guarantees.

Solvency proved in zero-knowledge

OBSERVATION

Each generation improved cryptographic sophistication.

What does Classic Merkle PoR verify?

Proof of Reserves tries to support one solvency claim: **`Total user liabilities` ≤ `CEX on-chain assets`**.

STEP 01

Inclusion check

QUESTION

"Is my balance included in the committed liability set?"

```
› verify_merkle_path(user_proof.json)
```

WHO

User

STEP 02

Root consistency check

QUESTION

"Am I checking against the same committed dataset as everyone else?"

```
› verify_merkle_root(snapshot_root.json)
```

WHO

User

STEP 03

Solvency check

QUESTION

"Are total user liabilities covered by exchange reserves?"

Auditor verifies privately

WHO

Auditor (with private record access)

TWO BOTTLENECKS

Users must participate to detect omission. Auditors are needed for aggregate solvency.

Where each PoR design pays its cost

Merkle pays in trust, **zk** pays in opacity, **LPOR** pays in size.

	GENERATION 2 Classic Merkle PoR	GENERATION 3 zk-PoR	THIS WORK LPOR
OMISSION DETECTION	User participation	User participation	Easy user participation
AGGREGATE SOLVENCY	Internal auditor	Cryptographic circuit	Public sum
MAIN LIMITATION	Needs internal auditor User runs script	Technically complex User runs script	Larger proof size Partial privacy trade-off

USER'S COMMITMENT VIEW	`0x7a81…`
	`0x4c92…`
	`0xa117…`
		`C = 0x92ab…`
		`π = 0x7f19…`
			`0x7a81…` • `TBTC-1`
			`0x4c92…` • `TBTC-0.1`
			`0xa117…` • `TBTC-0.01`

Three system properties — three different things users see.

Usability is a security parameter.

A proof's effective strength depends on how many independent users can — and do — verify it.

STEP 01

Better usability

Verification a non-technical user can complete in seconds.

STEP 02

More participation

Independent verifiers grow from a handful to thousands.

OUTCOME

Higher fraud detection

Omissions exposed with high probability across users.

$$P_{detect} = 1 - (1 - p)^k$$

LPOR architecture: Public Liability Ledger + Layered Verification

Users see liabilities at the bottom. Auditors verify commitments at the top. Both anchored to one Merkle root.

AUDITOR LAYER

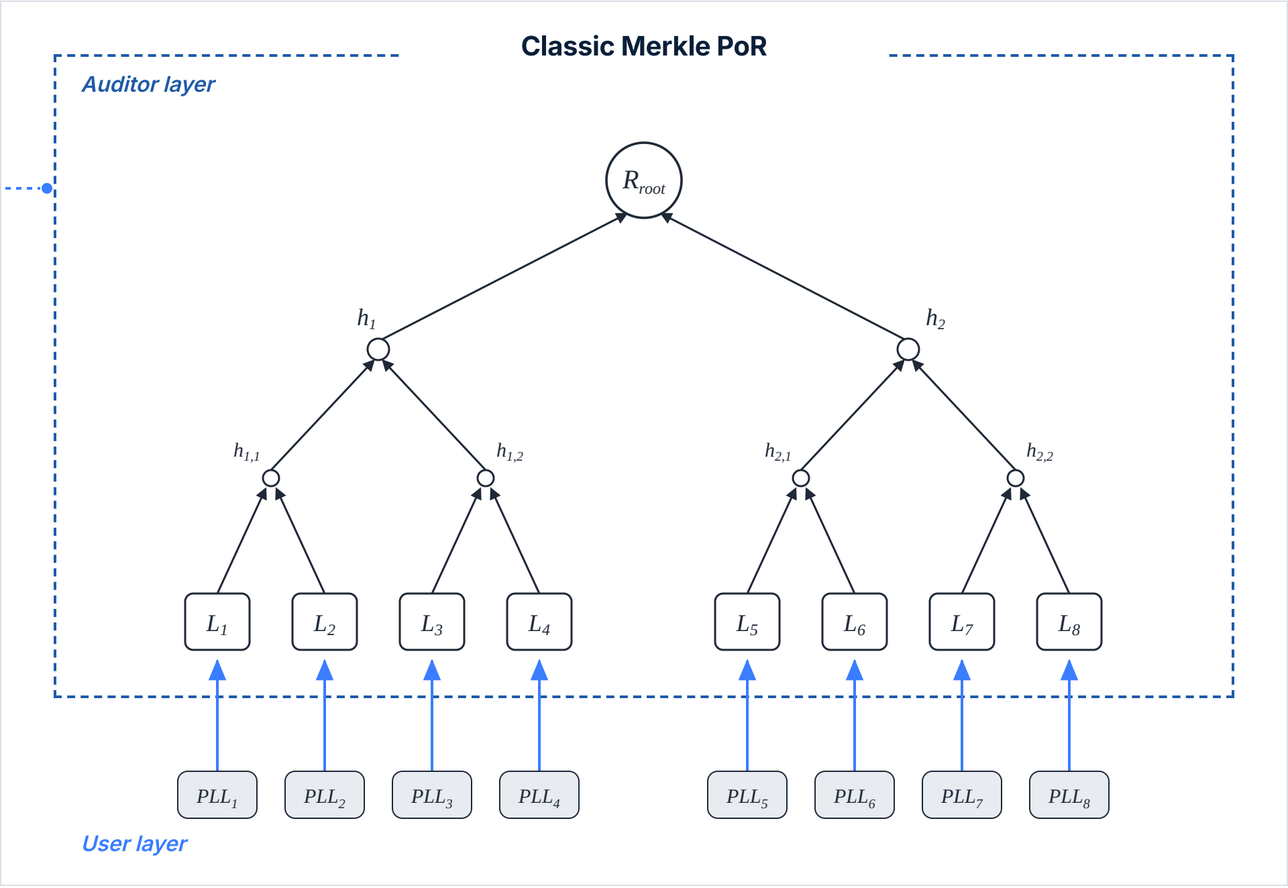
Cryptographic binding

- Verify Merkle leaves
- Verify Merkle root R_{root}
- Preserve binding end-to-end

PUBLIC LIABILITY LEDGER

Translation layer

- Human-readable liabilities
- Public summation Σ $Value_i$
- Deterministic $Leaf_i$ mapping



INSIDE THE DOTTED FRAME

Standard Merkle PoR

LPOR reuses the exact same Merkle commitment construction *above* the PLL row.

USER LAYER

Human verification

- Verify inclusion in PLL
- Verify total liabilities
- Verify reserve coverage

Users verify liabilities at the PLL row. Auditors verify commitments up the tree. Both meet at the same Merkle root.

From balance to Public Liability Ledger

Your balance becomes a few rows in a global, public, summable ledger.

STEP 1 · BALANCE TOKENIZER

1.12 BTC

Your balance at snapshot time

× 1	1 TBTC
× 1	0.1 TBTC
× 2	0.01 TBTC

Σ = 1.12 TBTC · your 4 rows

STEP 2 · PUBLIC LIABILITY LEDGER

Liability ledger · snapshot 2026-Q1 · excerpt

A few rows shown. The full ledger continues across all users.

USER	UUID	TOKEN	VALUE
— hidden —	0x1b27…aa01	TBTC-1	1.00
— hidden —	0xc59f…bd12	TBTC-1	1.00
YOU	0x9a4f…c021	TBTC-1	1.00
YOU	0xd5a2…01ef	TBTC-0.1	0.10
YOU	0x6f1b…44cd	TBTC-0.01	0.01
YOU	0x82e7…9a05	TBTC-0.01	0.01
	~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~		
— hidden —	0x4f2a…77b3	TBTC-10	100.00

AGGREGATE LIABILITY · ALL USERS

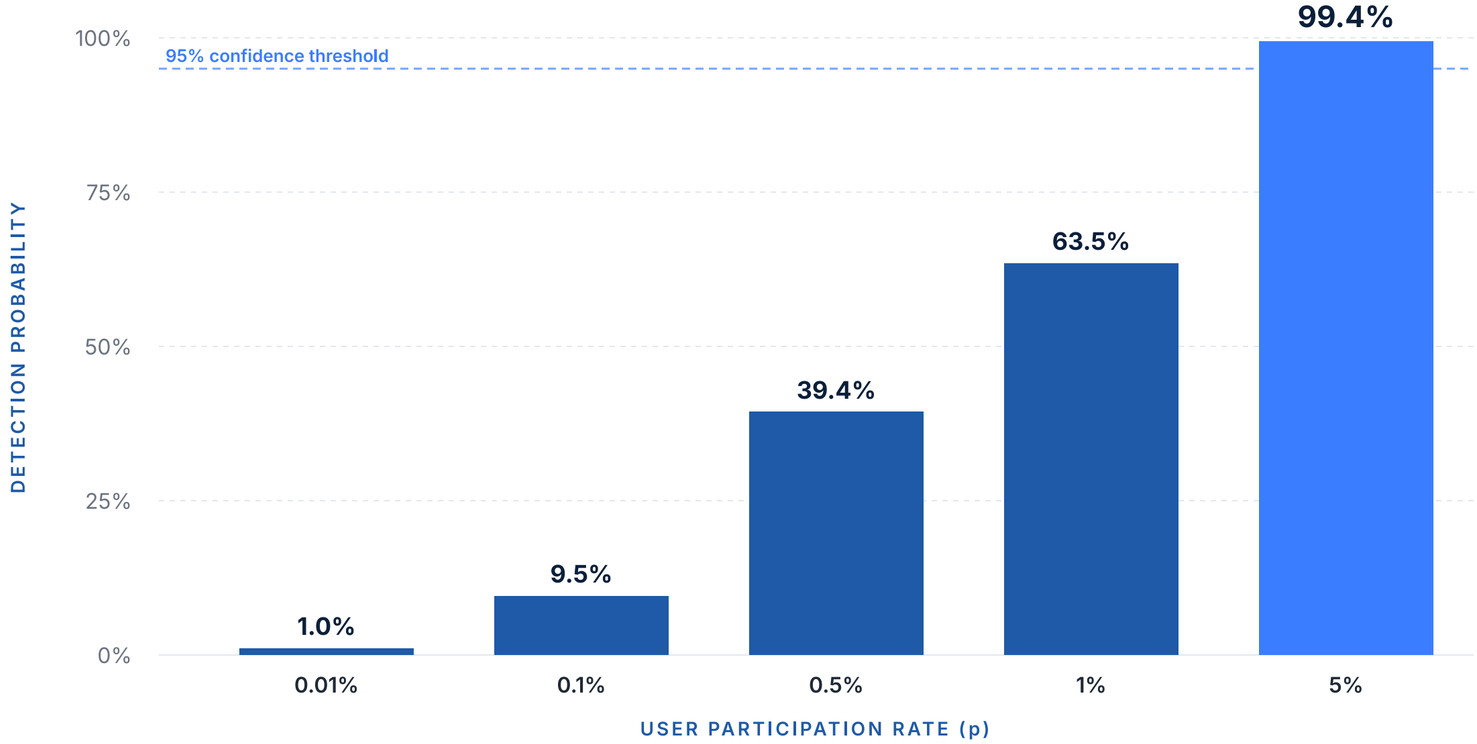
Σ = 1,247,058 TBTC

You see your rows. **Anyone** can sum the whole ledger.

09 · QUANTITATIVE RESULT

Better usability leads to better security

The cryptography does not change. Detection probability improves because participation increases.
Simulated: 10M users × 0.001% omission = **k = 100 omitted users**.



HEADLINE RESULT

99.4%

Omission detection probability

requires only

5%

of users to actually verify

MODEL

$$P_d = 1 - (1 - p)^k$$

Key takeaways

01
Classic Merkle PoR still relies on auditor-side aggregate liability verification.

02
zk-PoR removes auditor trust through cryptographic circuits.

03
LPOR removes auditor trust through public liability recomputation.

04
In PoR, usability can affect practical security through participation.

Cryptographic soundness is important.

But a proof that nobody verifies is not transparency.

Thank you · Questions?

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